

Rachna R

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PROFESSIONAL SUMMARY

AI & ML graduate with hands-on experience building end-to-end intelligent systems across computer vision, NLP, Generative AI, and multi-agent agentic architectures. Skilled in Python, TensorFlow, scikit-learn, LangGraph, RAG pipelines, and LLM-based technologies. Built and deployed real-world AI systems with measurable accuracy results. Passionate about building production-ready AI systems and rapidly learning emerging technologies.

EDUCATION

Avinashilingam University

Bachelor's Degree in Artificial Intelligence and Machine Learning | 2023 – 2026 | CGPA: 8.67/10

Government Higher Secondary School

Higher Secondary (12th) | 2023

Government Higher Secondary School

Secondary (10th) | 2021

TECHNICAL SKILLS

- **Programming Languages:** Python (Primary), JavaScript, HTML, CSS, SQL
- **ML Frameworks & Libraries:** TensorFlow, Keras, PyTorch (Fundamentals), scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **ML Concepts:** Supervised Learning, Unsupervised Learning, Neural Networks, CNNs, Decision Trees, KNN, Clustering, Regression, Classification
- **Model Evaluation:** Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix, Cross-Validation, Bias-Variance Tradeoff
- **AI & Generative AI:** NLP, Computer Vision, Generative AI, Prompt Engineering, Transfer Learning
- **Agentic AI & LLMs:** LangChain, LangGraph, Multi-Agent Systems, RAG, FAISS, Vector Databases, Groq AI, Ollama
- **Frameworks & Tools:** FastAPI, Streamlit, Flask, OpenCV, MediaPipe, YOLOv8, faster-whisper, Edge TTS
- **Tools & Platforms:** Git, GitHub, VS Code, Cursor, Jupyter Notebook, Google Colab
- **DevOps & Deployment:** REST API Development, CI/CD Pipelines, Version Control, Application Deployment

PROJECTS

Career Buddy – Multi-Agent AI Career Assistant

Python, LangGraph, LangChain, FastAPI, Streamlit, FAISS, Groq AI, faster-whisper, Edge TTS

- Architected a cooperative multi-agent system using LangGraph StateGraph with 5 specialized agents — Resume Analyst, Job Matcher, Interview Coach, Feedback Agent, and Roadmap Planner — orchestrated via conditional node routing.
- Built a semantic RAG pipeline using sentence-transformers and local FAISS vector stores for document ingestion and embedding-based retrieval.
- Integrated faster-whisper for Speech-to-Text and Edge TTS for voice synthesis, enabling a fully vocal interview simulation interface.
- Developed a wizard-style Streamlit frontend with glassmorphism UI and a headless FastAPI REST backend for full-stack deployment.

Vision-Based Virtual AI Security Guard System

Python, YOLOv8, DeepSort, Flask, OpenCV

- Built a real-time AI surveillance system achieving 91% human detection accuracy using YOLOv8 and DeepSort object tracking — evaluated using precision, recall and F1 metrics.
- Implemented automated alert system with adaptive learning capability and color-coded threat mechanisms, deployed as Flask web application.

Real-Time Hand Gesture Recognition and Adaptive Control System

Python, OpenCV, MediaPipe, KNN, scikit-learn

- Built a computer vision control system achieving 94% gesture classification accuracy using MediaPipe and KNN — involved data collection, feature engineering, model training and cross-validation evaluation.
- Implemented real-time mouse control and voice feedback with adaptive gesture classification.

Personal AI Assistant with NLP

Python, Flask, Ollama (Mistral)

- Developed a locally hosted NLP-based AI assistant achieving 88% intent recognition accuracy with zero external API dependency using Ollama (Mistral).
- Implemented speech recognition, intent classification, and system automation using Python NLP techniques.

INTERNSHIP

Generative AI & Prompt Engineering Intern | Gateway Software Solutions | Jun 2025 – Jul 2025

- Developed Generative AI workflows using Python, LLM APIs, and prompt engineering to improve model response accuracy.
- Designed optimized prompt templates and documented experimental results.

Python with Django Intern | Accent Techno Soft | Apr 2025 – May 2025

- Built backend modules using Python and Django for web application features and REST API endpoints.
- Assisted in application testing, bug fixing, and version management using Git.

CERTIFICATIONS

- Workshop on Generative AI, LLMs & Prompt Engineering – Accent Techno Soft
- Design Thinking & Agentic AI Bootcamp – SNS Institutions
- Gen AI Engineering Mastermind – Outskill